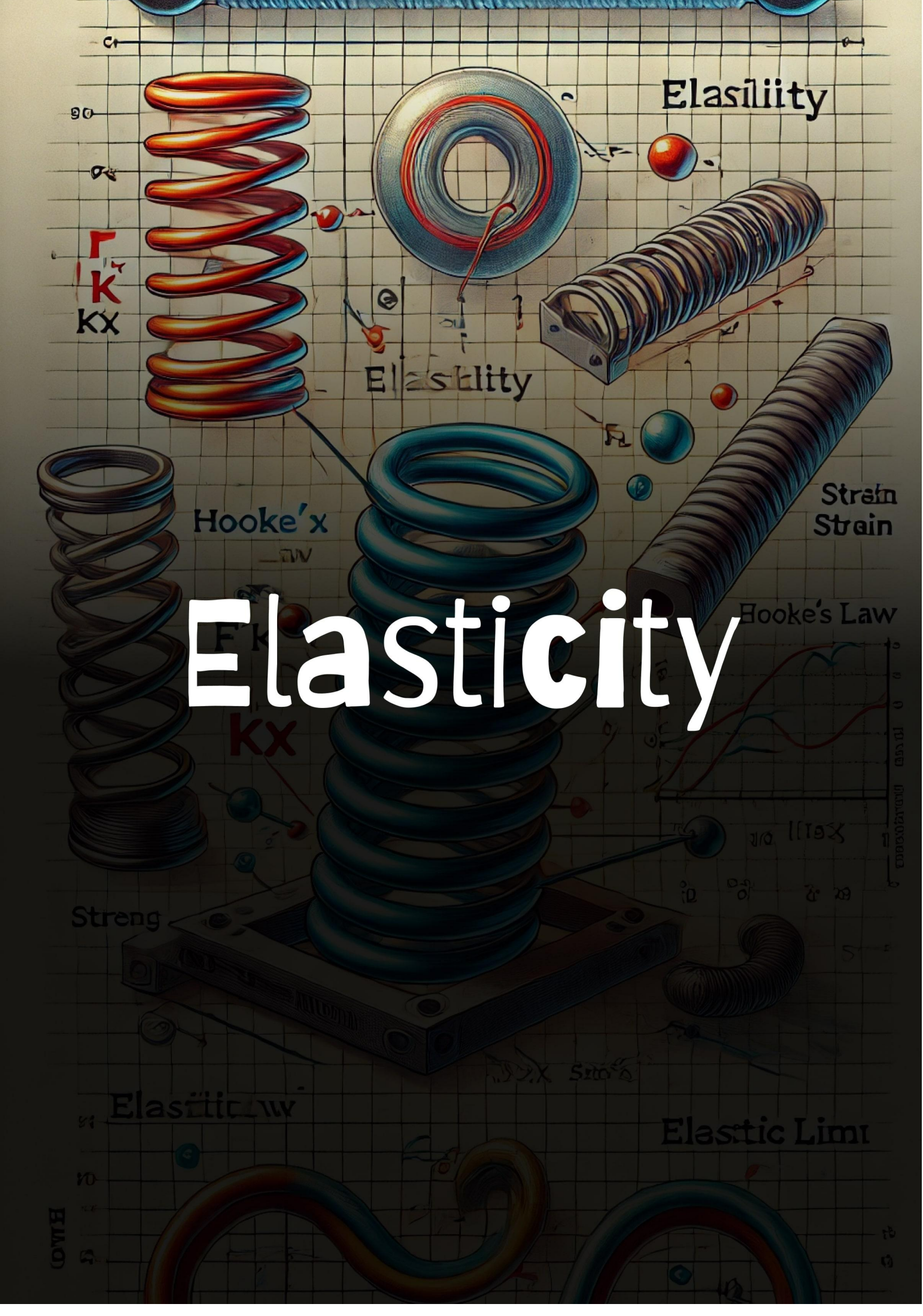


Elasticity

Elasticity

LONGITUDINAL STRESS (σ)

If F be the tension along the length of the rod whose cross sectional area is A, then we define longitudinal stress as

$$\sigma = \frac{F}{A}$$

There are two types of stress:
(i) Engineering stress
(ii) True stress

STRAIN (ϵ)

If an element of original length dx elongates by a length dy due to longitudinal stress, then we define local strain as

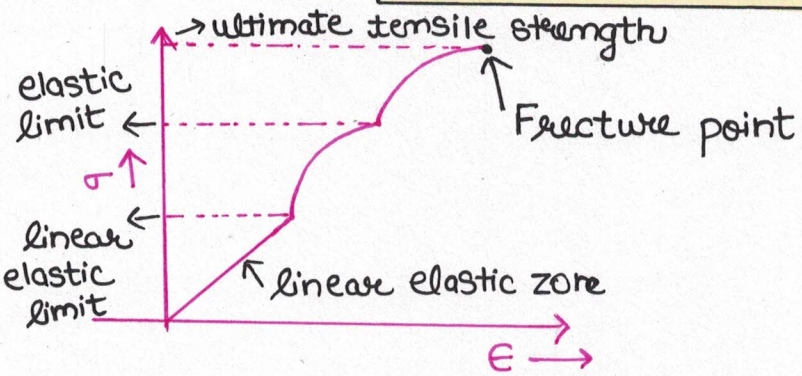
$$\epsilon = \frac{dy}{dx}$$

If all the elements of a rod have same strain then we can also write

$$\epsilon = \frac{\Delta l}{l}$$

where Δl is change in length and l is original length.

STRESS Vs STRAIN (GRAPH)



YOUNG'S MODULUS (Y)

The ratio of stress to strain in the linear elastic limit is called young's modulus.

$$Y = \frac{\sigma}{\epsilon}$$

$$Y = \frac{Fl}{A\Delta l}$$

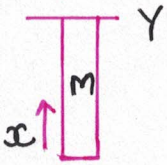
250

Que: A rod of mass M & length ℓ (unstretched) hang from a ceiling. Find

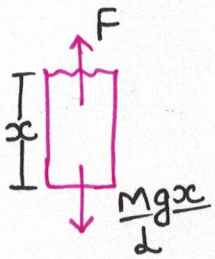
(i) Tension as a function of distance from the bottom

(ii) σ & ϵ as a function of distance x

(iii) Total elongation of rod.



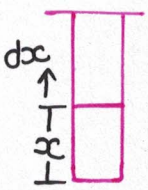
(i) $F = \frac{Mgx}{\ell}$



(ii) $\sigma = \frac{Mgx}{\ell A}$

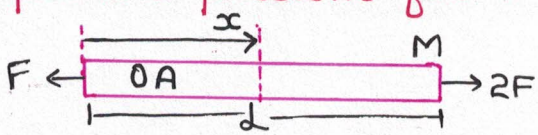
$$\epsilon = \frac{Mgx}{\ell AY} = \frac{dy}{dx}$$

(iii) $\int_0^{\Delta\ell} dy = \frac{Mg}{\ell AY}\int_0^{\ell} x.dx$

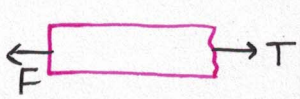


$$\Delta\ell = \frac{Mg\ell}{2AY}$$

Que: Repeat the previous problem for the shown situation.



$$T - F = \frac{Mx}{\ell}.\frac{F}{M}$$



$$T = F\left(\frac{x}{\ell}+1\right)$$

$$\sigma = \frac{F}{\ell A}(x+\ell)$$

$$\epsilon = \frac{F}{\ell AY}(\ell+x) = \frac{dy}{dx}$$

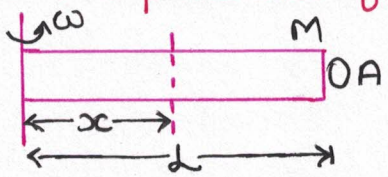
$$\int_0^{\Delta\ell} dy = \int_0^{\ell} \frac{F}{\ell AY}(\ell+x).dx$$

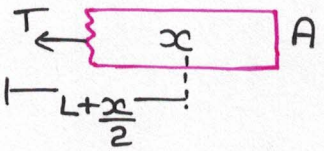
$$\Delta\ell = \frac{F}{\ell AY}\left[\ell x + \frac{x^2}{2}\right]_0^{\ell}$$

$$\Delta\ell = \frac{3}{2}\frac{F\ell}{AY}$$

251

Que.) Repeat the previous problem for the shown situation.





$$T = \frac{\omega^2 (L+x)}{2} \times \frac{M(L-x)}{L}$$

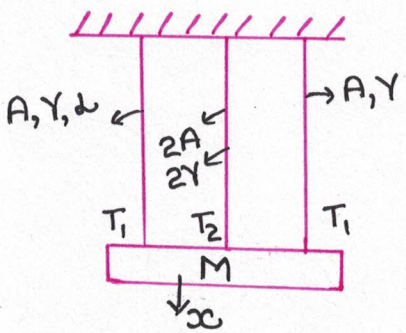
$$T = \frac{M\omega^2}{2L} (L^2 - x^2)$$

$$\int_0^{\Delta L} dy = \frac{M\omega^2}{2LAY} \int_0^{L} (L^2 - x^2) \cdot dx$$

$$\Delta L = \frac{M\omega^2}{2LAY} \left(L^3 - \frac{L^3}{3}\right)$$

$$= \frac{M\omega^2 L^2}{3AY}$$

Que.) A block of mass M hangs by 3 wires, two of them being identical as shown. Find the tensions T_1 and T_2 and also the elongation x.



$$Y = \frac{FL}{A\Delta L}$$

$$F = \frac{AY\Delta L}{L}$$

$$F = \frac{AYx}{L}$$

$$T_1 = \frac{AYx}{L}$$

$$T_2 = \frac{4YxA}{L}$$

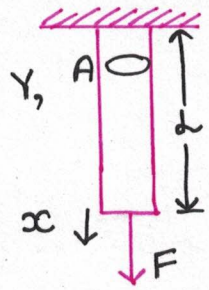
$$\frac{2AYx}{L} + \frac{4AYx}{L} = Mg$$

$$x = \frac{Mg.L}{6AY}$$

$$T_1 = \frac{Mg}{6} \quad , \quad T_2 = \frac{2Mg}{3}$$

252

SPRING EQUIVALENT OF A MASSLESS ELASTIC ROD





$$Y = \frac{F\ell}{Ax}$$

$$F = \frac{YAx}{\ell} = K_{eff}.x$$

$$\boxed{K_{eff} = \frac{YA}{\ell}}$$

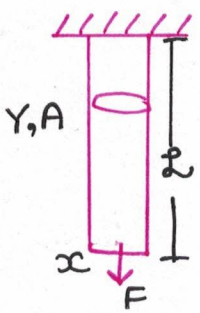
ELASTIC ENERGY DENSITY IN A ROD

$$u = \frac{1}{2} \times \left(\frac{YA}{\ell}\right) x^2$$

$$V = A\ell \quad \text{(volume)}$$

$$\mu = \frac{u}{V} = \frac{1}{2}\frac{YA}{\ell}.x^2 \times \frac{1}{A\ell}$$

$$= \frac{1}{2} Y \left(\frac{x}{\ell}\right)^2$$



$$\mu = \frac{1}{2} \times Y \times \epsilon^2 \qquad (\mu \rightarrow \text{energy per unit volume})$$

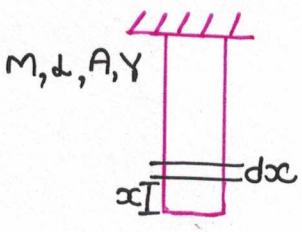
$$Y = \frac{\sigma}{\epsilon}$$

$$\mu = \frac{1}{2}\sigma\epsilon$$

$$\mu = \frac{1}{2}\frac{\sigma^2}{Y}$$

Que.) Find elastic energy stored in the rod due to elongation under its own weight.

$$\epsilon = \frac{Mgx}{\ell AY}$$



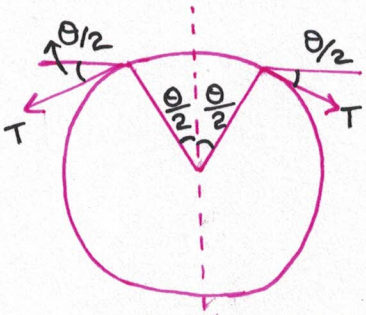
$$du = \frac{1}{2} \times Y \times \left(\frac{Mgx}{\ell AY}\right)^2$$

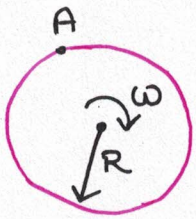
$$\int_0^u du = \frac{M^2g^2}{2\ell^2 AY} \int_0^\ell x^2.dx$$

$$u = \frac{M^2g^2\ell}{6AY}$$

253

Que: Find (i) The tension in the ring as a function of ω & (ii) Maximum permissible ω so that ring does not break.





$$2T \sin\left(\frac{\theta}{2}\right) = \rho (AR\theta)\omega^2 R$$

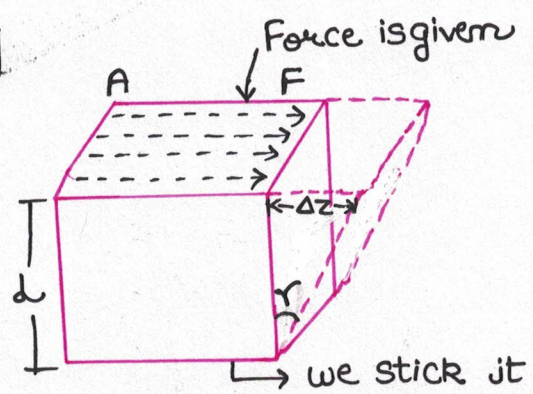
$$T\theta = \rho A \omega^2 R^2 \theta$$

$$T = \rho A \omega^2 R$$

$$\sigma = \frac{T}{A} = \rho\omega^2 R^2 < \sigma_0$$

$$\omega < \sqrt{\frac{\sigma_0}{\rho R^2}}$$

SHEAR STRESS



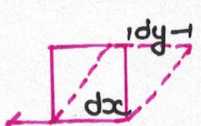
If a force F acts parallel to a surface area A, then we define the shear stress as

$$\left(\sigma_s = \frac{F}{A}\right)$$

SHEAR STRAIN

Let the upper edge of the shown block get displaced through a distance ΔL more, then we define shear strain as

$$E_s = \frac{\Delta L}{L} = \tan\gamma$$



$$E_s = \frac{dy}{dx}$$

where L is the vertical thickness of the block.

SHEAR MODULUS

The ratio of shear stress to shear strain is called shear modulus.

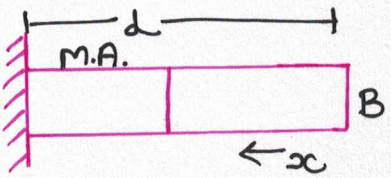
$$\eta = \frac{\sigma_s}{E_s}$$

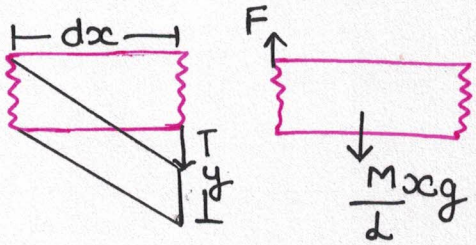
Que. A candylever beam comes out of the wall as shown in figure. Find

(i) shear stress as a function of x.

(ii) shear strain as a function of x.

(iii) Amount by which end B comes down.





$$F = \frac{Mxg}{l}$$

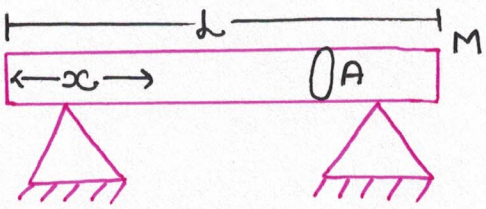
$$\sigma_s = \frac{F}{A} = \frac{Mgx}{lA}$$

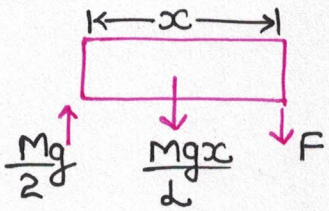
$$\epsilon_s = \frac{\sigma}{\eta} = \frac{Mgx}{lA\eta} = \frac{dy}{dx}$$

$$\int_0^{\Delta l} dy = \frac{Mg}{lA\eta} \int_0^{l} x.dx$$

$$\Delta l = \frac{Mgl}{lA\eta}$$

Que. Find shear strain and shear stress.





$$F = \frac{Mg}{2} - \frac{Mgx}{l}$$

$$F = Mg\left[\frac{1}{2} - \frac{x}{l}\right]$$

$$\sigma_s = \frac{F}{A} = \frac{Mg}{A}\left[\frac{1}{2} - \frac{x}{l}\right]$$

Que. Find the max. displacement in the previous que.

$$\epsilon = \frac{\sigma_s}{\eta}$$

$$\epsilon = \frac{Mg}{2A\eta}\left[1 - \frac{2x}{l}\right] = \frac{dy}{dx}$$

$$\int_0^{\Delta l} dy = \left[\frac{Mgx}{2A\eta} - \frac{x^2 Mg}{2lA\eta}\right]_0^{l/2}$$

$$\Delta l = \frac{Mgl}{4A\eta} - \frac{lMg}{A\eta 8} = \frac{Mgl}{8A\eta}$$

255

BULK STRESS

Let a substance be surrounded by an atmosphere of pressure P and let dP be the change in pressure, then the bulk stress is defined as

$$\sigma_B = dP$$

BULK STRAIN

Let dV be change in volume of a substance having original volume V. Then we define bulk strain as

$$\epsilon_B = \frac{dV}{V}$$

BULK MODULUS

The magnitude of ratio of bulk stress to bulk strain is called bulk modulus.

$$B = \frac{-dP}{dV/V} = \frac{-\sigma_B}{\epsilon_B}$$

Que.) Find the bulk modulus of an ideal gas for an adiabatic process.

$$PV^\gamma = C$$

$$\ln P + \gamma \ln V = \ln C$$

$$\frac{1}{P}\frac{dP}{dV} + \frac{\gamma}{V} = 0$$

$$\frac{-dP}{dV/V} = \gamma P$$

★ For laboratory purposes, we can write,

$$B = \frac{-\Delta P}{(\Delta V/V)}$$

BULK MODULUS IN TERMS OF DENSITY

$$\rho = \frac{m}{V}$$

$$\ln \rho = \ln m - \ln V$$

$$\frac{1}{\rho}\frac{d\rho}{dV} = -\frac{1}{V} \qquad (\text{w.r.t. } V)$$

$$\frac{-dV}{V} = \frac{d\rho}{\rho}$$

$$\boxed{B = \frac{dP}{(d\rho/\rho)}}$$

256

Que: The density on the surface of well is ρ_0, and bulk modulus is B_0 (assume constant). Find relation b/w depth & density.

$$P = P_0 + \rho g y$$

$$dP = \rho g\, dy$$

$$B_0 = \frac{dP}{(d\rho/\rho)}$$

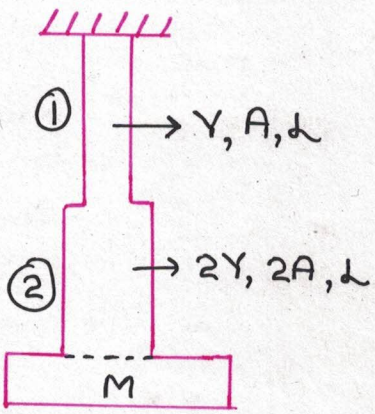
$$\frac{B_0}{\rho} = \frac{\rho g\, dy}{d\rho}$$

$$\int_{\rho_0}^{\rho} \frac{B_0}{\rho^2 g}\, d\rho = \int_0^d dy$$

$$\left[\frac{-B_0}{\rho g}\right]_{\rho_0}^{\rho} = [y]_0^d$$

$$\frac{gy}{B_0} = -\frac{1}{\rho} + \frac{1}{\rho_0}$$

Que: Find the individual elongations and total elongation. Also find K_{eff} of the combination.



$$T = Mg$$

$$\sigma_T = \frac{Mg}{A}$$

$$\epsilon_1 = \frac{Mg}{AY} = \frac{x_1}{\ell}$$

$$x_1 = \frac{Mg\ell}{AY}$$

Similarly, $x_2 = \frac{Mg\ell}{4AY}$

$$x = x_1 + x_2 = \frac{5Mg\ell}{4AY}$$

$$K_{eff} = \frac{Mg}{x}$$

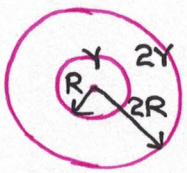
$$= \frac{4AY}{5\ell}$$

COMPRESSIBILITY

The reciprocal of ~~block~~ bulk modulus is called compressibility.

$$K = \frac{1}{B} = \frac{-\epsilon_B}{\sigma_B}$$

Que.) Find the elongation in the cable and elastic energy stored in the cable.



$$K_1 = \frac{Y\pi R^2}{L}$$

$$\therefore K_2 = \frac{2Y.3\pi R^2}{L} = \frac{6Y\pi R^2}{L}$$

$$K_{eff} = K_1 + K_2$$

$$= \frac{7Y\pi R^2}{L} = \frac{Mg}{x}$$

$$x = \frac{MgL}{7Y\pi R^2}$$

$$U = \frac{1}{2} K_{eff} \cdot x^2$$

$$U = \frac{1}{2}\left(\frac{Mg}{x}\right)x^2$$

$$U = \frac{Mgx}{2} = \frac{M^2g^2L}{14YAR^2}$$